4.6 The Fundamental Theorem of Algebra

Name _____ Period ____ Date _____

Algebra II 4.6 (30-minute plan) · Student sheet

I can…

- use the Fundamental Theorem of Algebra to find the number of solutions of a polynomial equation.
- find all the zeros of a polynomial function and write polynomials from their zeros.
- use Descartes's Rule of Signs.

Vocabulary · fill in the missing word

Fundamental Theorem of Algebra — a _____ equation of degree n > 0 has exactly n solutions, counting repeated and imaginary solutions

Complex Conjugates Theorem — if a + bi is a zero of a polynomial with real _____, so is a − bi

Descartes's Rule of Signs — the number of _____ real zeros is the number of sign changes in f(x) or less than that by an even number; use f(−x) for negative zeros

Part 1 · Notes and examples (fill in each note, then work the example)

Fundamental Theorem of Algebra: A polynomial equation of degree n has exactly _____ solutions.

Example 1

How many solutions does $x^3 + 64x = 0$ have? Find them.

Complex Conjugates Theorem: Imaginary zeros come in _____ pairs: a + bi and _____.

Descartes's Rule of Signs: Descartes's Rule: count the sign changes in _____ and in _____.

Example 2

Write a polynomial function f of least degree with leading coefficient 1 that has the zeros 3 and 2i.

Part 2 · Partner practice

Partner practice

Write a polynomial function of least degree with leading coefficient 1 that has the zeros 1 (repeated) and 4i.